



GraSTI-ACL: Graph spatial–temporal infomax with adversarial contrastive learning for brain disorders diagnosis based on resting-state fMRI

Biao He^{a,b}, Erni Ji^c, Xiaofen Zong^d, Zhen Liang^{a,b}, Gan Huang^{a,b}, Li Zhang^{a,b} *

^a School of Biomedical Engineering, Medical School, Shenzhen University, Shenzhen, 518060, Guangdong, China

^b Guangdong Provincial Key Laboratory of Biomedical Measurements and Ultrasound Imaging, Shenzhen, 518060, Guangdong, China

^c Department of Bipolar Disorder, Shenzhen Kangning Hospital, Shenzhen Mental Health Center, Shenzhen, 518020, Guangdong, China

^d Department of Psychiatry, Renmin Hospital of Wuhan University, Wuhan, 430060, Hubei, China

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ABSTRACT

Resting-state functional magnetic resonance imaging (rs-fMRI) has been widely used in research on brain disorders due to its informative spatial and temporal resolution, and it shows growing potential as a non-invasive tool for assisting clinical diagnosis. Among various methods based on rs-fMRI, graph neural networks have received significant attention because of their inherent structural similarity to functional connectivity networks (FCNs) of the brain. However, constructing FCNs that effectively capture both spatial and temporal information from rs-fMRI remains challenging, as traditional methods often rely on static, fully connected graphs that risk redundancy and neglect dynamic patterns. Based on the information bottleneck principle, this paper proposes a graph augmentation strategy named Graph Spatial–Temporal Infomax (GraSTI) to adaptively preserve both global spatial brain-wide FCNs and local temporal dynamics. We integrate GraSTI with theoretical explanations and design a practical implementation to adapt to our graph augmentation strategy and enhance feature capture capability. Furthermore, GraSTI is incorporated into an adversarial contrastive learning framework to achieve a mutual information equilibrium between graph representation effectiveness and robustness for downstream brain disorders diagnosis tasks. The proposed method is evaluated on datasets from three different brain disorders: Alzheimer's disease (AD), major depressive disorder (MDD), and bipolar disorder (BD). Extensive experiments demonstrate that the proposed GraSTI-ACL achieves diagnostic accuracy gains of 0.13% to 23.56% for AD, 1.23% to 13.81% for MDD, and 2.53% to 24.53% for BD diagnosis over existing methods. Meanwhile, our method demonstrates strong interpretability in identifying relevant brain regions and connectivities for different brain disorders.

1. Introduction

The study of brain diseases using resting-state functional magnetic resonance imaging (rs-fMRI) has become a crucial area in neuroimaging research. Rs-fMRI captures blood oxygen level-dependent (BOLD) signals, which provide rich spatial and temporal information in a non-invasive manner. These signals contain valuable information related to brain function, cognition, and behavior (Jbabdi et al., 2015; Betzel and Bassett, 2017; Li et al., 2020, 2019; Wang et al., 2021). Though its clinical application remains primarily exploratory rather than standard practice, rs-fMRI has shown promise in elucidating functional connectivity alterations associated with various brain disorders.

Among various analytical methods, machine learning approaches based on graph neural networks (GNNs) have gained significant attention. This is largely due to the inherent similarity between functional connectivity networks (FCNs) of the brain and graph structures (Timón-Reina et al., 2021; Wang et al., 2022; Liu et al., 2023). Typically, an

FCN is represented as a graph where each node corresponds to a brain region of interest (ROI), and edges represent the correlations between pairs of ROIs (Zhang et al., 2024). Several approaches, such as (Li et al., 2021; Thapaliya et al., 2025a, 2024), have introduced ROI-aware aggregation mechanisms to uncover shared brain region clustering patterns across subjects. While these methods help alleviate biases introduced by strong prior assumptions and hold potential for revealing biologically meaningful brain organization, they do not effectively leverage the dynamic information inherent in rs-fMRI signals. Despite the powerful embedding capabilities of graphs, constructing effective graph representations that fully leverage the information contained in the rs-fMRI data remains a significant challenge.

With the success of contrastive learning in natural image classification, graph contrastive learning (GCL) has emerged as a prominent research focus in recent years (Zhang et al., 2023). In this paradigm, the learning algorithm acquires representations by contrasting augmented

* Corresponding author at: School of Biomedical Engineering, Medical School, Shenzhen University, Shenzhen, 518060, Guangdong, China.
E-mail address: lzhang@szu.edu.cn (L. Zhang).

graph variants as positive samples against other instances as negative samples. However, many approaches using rs-fMRI data often model the brain FCN topology through static and fully-connected weighted adjacency matrices that retain redundant information. Meanwhile, the topology remains fixed throughout the learning process (Ktena et al., 2018). Unlike natural image processing scenarios, FCN nodes corresponding to brain ROIs inherently possess multidimensional attributes, including anatomical spatial coordinates, neural activation patterns, and temporal dynamics. Some recent studies incorporated amplitude of low-frequency fluctuations (ALFF) as nodal features due to their ability to encode physiological cognitive states across different experimental paradigms (Chen et al., 2023a). Also, the inherent temporal nature of BOLD signals further necessitates explicit modeling of FCN dynamic properties during graph construction (Bullmore and Sporns, 2009). Conventionally, the sliding-window approach introduces dynamic information by constructing a separate graph for each time window and then connecting them to form a large-scale dynamic temporal graph (Liu et al., 2025; Kim et al., 2021). While this strategy can capture temporal variations, it often brings substantial redundancy and enlarges the input scale. In practice, this redundancy usually requires models with higher representational capacity (i.e., more parameters) to process the expanded graph sequence, which may obscure meaningful neural patterns and complicate model training (Chen et al., 2023b; Xu et al., 2023).

Another aspect that needs to be taken into consideration is the heterogeneity among different brain disorders. For example, Alzheimer's disease (AD) is a neurological disorder that often leads to abnormalities in both brain structure and function (Dang et al., 2023). In contrast, major depressive disorder (MDD) and bipolar disorder (BD) are mental illnesses in which the brain's structure remains similar to that of healthy individuals, but brain function undergoes significant changes (Clark and Sahakian, 2008). The brain regions affected by these disorders also differ. Therefore, learning stable graph representations across different brain disorders to derive FCN topology from spatially and temporally informative rs-fMRI data, while maintaining a balance between information necessity and sufficiency, is the primary issue to be addressed.

Inspired by the information bottleneck (IB) principle (Tishby et al., 2000), this paper introduces *Graph Spatial-Temporal Infomax with Adversarial Contrastive Learning* (GraSTI-ACL) for rs-fMRI-based brain disorder diagnosis. Our solution comprises three major components. First, we propose a graph augmentation method that infers FCN topologies by integrating global spatial and local temporal information of ROIs. Node features are extracted from multi-band ALFF, while edge weights are derived from static global functional connectivity (FC) in BOLD time series. To capture temporal dynamics, we use multiple time windows to capture the non-stationarity property of rs-fMRI. An adversarial info-minmax strategy approximates the information bottleneck, balancing necessity and sufficiency while reducing computational load. Constraints on the upper and lower bounds of information are imposed to regulate redundant information within a reasonable range. Second, we incorporate dynamic temporal information as an augmentation to the existing info-minmax process. To improve graph representations, we introduce a newly designed topology attention encoder (TAE), which balances topological integrity and global attention. Together, these components form GraSTI, a graph augmentation strategy for generating positive graph samples. Third, we integrate GraSTI with an adversarial contrastive learning (ACL) framework to enhance graph representations. An additional constraint ensures the augmented graph retains essential information and remains stable across different augmentation levels. This design balances mutual information, optimizing effectiveness and robustness for brain disorder diagnosis.

To validate the performance of our algorithm, we applied the proposed GraSTI-ACL to the diagnosis of three types of brain disorders: the neurodegenerative disease AD, the psychiatric disorder MDD and BD. We conducted a comprehensive comparison with several current mainstream methods.

In summary, we make the following contributions:

- We theoretically extend the IB principle to graph-based learning to control redundant information.
- We incorporate spatial and temporal information from time series into a graph augmentation strategy.
- We design a node feature encoding framework for a better graph representation.

The rest of the paper is organized as follows. We first provide the motivation based on the IB principle and the necessary preliminaries. Next, we describe the proposed method with detailed theoretical explanations. The experimental results are then presented. Finally, we conclude the paper. The codes implementing our method are available at: <https://github.com/BiaoHe2025/GraSTIACL>.

2. Preliminaries and notations

First, we introduce the notation used in this paper. A graph is denoted as $G(V, E, A, W)$, where $V \in \mathbb{R}^{N \times d}$, N is the number of graph nodes and d is dimension of node feature. The adjacency matrix is defined as $A \in \mathbb{R}^{N \times N}$, where $A_{ij} = 1$ indicates an FC between the i th and j th nodes. $E, W \in \mathbb{R}^{N \times N}$, where W is the attribute of the edge matrix E and is obtained by computing the Pearson correlation between the entire BOLD time series. W_{ij} represents the connectivity strength between the i th and j th nodes, also known as edge weights. It is worth noting that the temporal dynamic weights do not participate in the global FCN construction.

The graph is initially set up as a fully connected graph, with all elements in A initially set to 1. Our goal is to design a self-supervised encoder to preserve essential information from the existing node matrix V and edge weight matrix W . This process can be viewed as graph augmentation, transforming a graph G into an augmented graph denoted as G' .

Next, we introduce the theoretical motivation behind the proposed method. In information theory, for two random variables X and Y , their mutual information I is defined as

$$I_{\Theta}(X, Y) = \int p(X, Y) \log \frac{p(X, Y)}{p(X)p(Y)} dX dY, \quad (1)$$

where p is the prior distribution. We can understand mutual information as a measure of how much information two variables share with each other. Consider a Markov chain $X \rightarrow Z \rightarrow Y$, where X is the original sample, Z is the representation obtained through the encoder, and Y is the label of the sample inferred from Z . Intuitively, the more information X contains about the sample label Y (i.e., the larger $I(X, Y)$), the more likely this chain points in the correct direction without requiring a complicated encoder, or even without an encoder at all. However, in reality, X often contains a significant amount of information that is irrelevant to Y , implying that useful information has an upper limit, known as the information bottleneck (Tishby et al., 2000). Therefore, the purpose of Z is to retain more information relevant to the downstream task Y while discarding irrelevant information, thereby improving the performance of our model on the downstream task. Consequently, we have the following objective function to train an effective encoder:

$$\max IB = I(Z, Y) - \beta I(Z, X), \quad (2)$$

where β is to control the threshold of reserved information.

By the definition of mutual information in (1), we can further derive a specific objective function. In deep learning, variational information bottleneck (VIB) (Alemi et al., 2016) defines the IB objective function through a variational approximation. However, it requires downstream labels for optimization. Based on graph contrastive learning, adversarial-GCL (AD-GCL) no longer preserve mutual information from the labels but instead uses the augmented representation of the graph itself as positive samples (Suresh et al., 2021). Its objective function is

$$\min_{\Phi} \max_{\Theta} I(f_{\Theta}(G), f_{\Theta}(t_{\Phi}(G))), \quad (3)$$

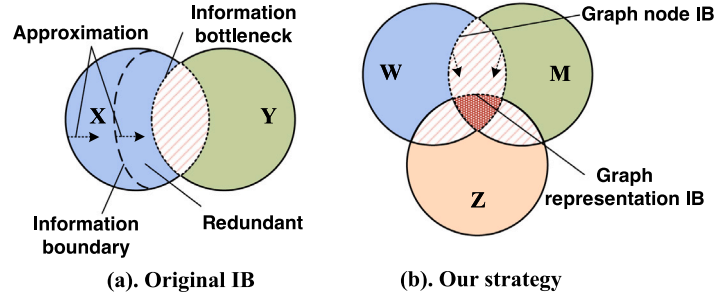


Fig. 1. An illustration of our motivation by comparison (a) original IB to (b) our proposed strategy. In our strategy, we first obtain an augmentation via graph node IB and then encode the graph via graph representation IB.

where $t_\Phi(G)$ denotes a graph data augmentation of G , Φ corresponds to the learnable parameters of the augmenter GNN and multilayer perceptron (MLP), and Θ corresponds to the learnable parameters of the GNN f . This approach learns graph representations without relying on downstream labels and the edge dropout is guided by the graph representation to train an MLP using concatenated node vectors as input. However, this method makes it difficult to determine whether redundant information in the node vectors has been sufficiently discarded.

To better illustrate the proposed idea, Fig. 1 shows the principles of the original IB and our strategy. Fig. 1(a) illustrates the original IB principle, where the input X undergoes optimization to push its information boundary toward the theoretical information bottleneck. This process removes redundant information while learning representations more relevant to the label Y . In contrast, our strategy first perform graph node IB by leveraging both the edge weight matrix W and the nodal feature inner product matrix M , which incorporates node information and eliminates redundancy. Subsequently, inspired by AD-GCL (Suresh et al., 2021), we construct a network capable of learning more robust features at the level of graph representation Z to adapt to different brain disorders diagnosis scenarios via graph representation IB. As illustrated in Fig. 1(b), we first obtained an augmented graph via graph node IB. Then, through adversarial graph contrastive learning, the augmentation method (edge dropout) is optimized to preserve more effective information via graph representation IB.

3. Methodology

In this section, we present a detailed theoretical explanation of our methodology. Our approach can be summarized in two main parts: the first is the graph spatial-temporal infomax, including our designed topology attention encoder, and the second is the ACL framework. The entire structure is illustrated in Fig. 2.

3.1. Original graph construction

The construction of the original graph is illustrated in Fig. 2. Each ROI is treated as a node in the graph. The FC between each pair of ROIs is computed using the Pearson correlation coefficient (PCC) of their BOLD signals, reflecting the connection between the corresponding brain regions. Thus, FC is used as the edge weight matrix. Node features are initialized using ALFF features from three selected frequency bands. Combining these elements yields the original graph $G(V, E, A, W)$. Additionally, we divide the BOLD signal time series into multiple non-overlapping segments with a length of 40 repetition times (TR) and compute local PCCs to capture temporal dynamics. Furthermore, an adjacency matrix with all entries equal to one, indicating a fully connected initial functional brain network, is used for graph augmentation.

3.2. Graph spatial-temporal infomax

Resting-state fMRI data, while highly informative, lacks a well-defined graph structure. Traditional approaches typically construct FCNs and directly apply to end-to-end models for downstream tasks. However, this approach risks introducing redundant information that may lead to unsatisfactory graph representations. To address this issue, we propose a node IB-guided framework that selectively filters and retains necessary information to learn the graph topology.

Specifically, for the original graph $G(V, E, A, W)$, our goal is to maximize the following function:

$$\max I_\Theta(A_{ij}, M_{ij}) - \beta I_\Theta(A_{ij}, W_{ij}), \quad (4)$$

where A_{ij} represents the FC between the i th and j th nodes, W_{ij} is the corresponding edge weights, $M_{ij} = \sigma(\mathbf{v}_i \cdot \mathbf{v}_j)$ is the dot product of the i th and j th nodal feature vectors $\mathbf{v}_i$ and $\mathbf{v}_j \in \mathbb{R}^d$ passed through a sigmoid activation function $\sigma(\cdot)$, β is a hyperparameter controlling the information upper bound of the adjacency matrix. The node feature vectors, composed of ALFF features from three frequency bands, represent brain intrinsic state information. Thus, the first term of (4) maximizes the mutual information between the adjacency matrix and the node spatial information. To regulate the amount of preserved information and eliminate redundancies, the second term minimizes the mutual information between the adjacency matrix and the edge weight matrix representing temporal information. By maximizing (4), we aim to update the fully connected initial adjacency matrix of the original graph via graph node IB, making it more relevant to the node information and obtaining an augmented graph that balances information sufficiency and necessity.

To compute the first term, we introduce the probability distribution $p(A_{ij}, M_{ij})$, defined by our encoder and the Markov chain ($M_{ij} \leftrightarrow W_{ij} \leftrightarrow A_{ij}$). Assuming the variational approximation $q(M_{ij} | A_{ij})$ for $p(M_{ij} | A_{ij})$ and using the mutual information definition from the previous section, we obtain the variational lower bound:

$$\begin{aligned} I_\Theta(A_{ij}, M_{ij}) &\geq \int p(A_{ij}, M_{ij}) \log \frac{q(M_{ij} | A_{ij})}{p(M_{ij})} dA_{ij} dM_{ij} \\ &= \int p(A_{ij}, M_{ij}) \log q(M_{ij} | A_{ij}) dA_{ij} dM_{ij} \\ &\quad - \int p(A_{ij}, M_{ij}) \log p(M_{ij}) dA_{ij} dM_{ij}. \end{aligned} \quad (5)$$

Noting that the second term of Eq. (5) is the entropy of the node vector dot product and is irrelevant to our optimization objective, it can thus be omitted. In this way, Eq. (5) becomes:

$$I_\Theta(A_{ij}, M_{ij}) \geq \int p(A_{ij}, M_{ij}) \log q(M_{ij} | A_{ij}) dA_{ij} dM_{ij}. \quad (6)$$

For the second term of (4), assuming a Gaussian distribution $r(A_{ij})$ as a variational approximation of the true distribution, we obtain the variational upper bound:

$$I_\Theta(A_{ij}, W_{ij}) \leq \int p(A_{ij}, W_{ij}) \log \frac{p(A_{ij} | W_{ij})}{r(A_{ij})} dA_{ij} dW_{ij}. \quad (7)$$

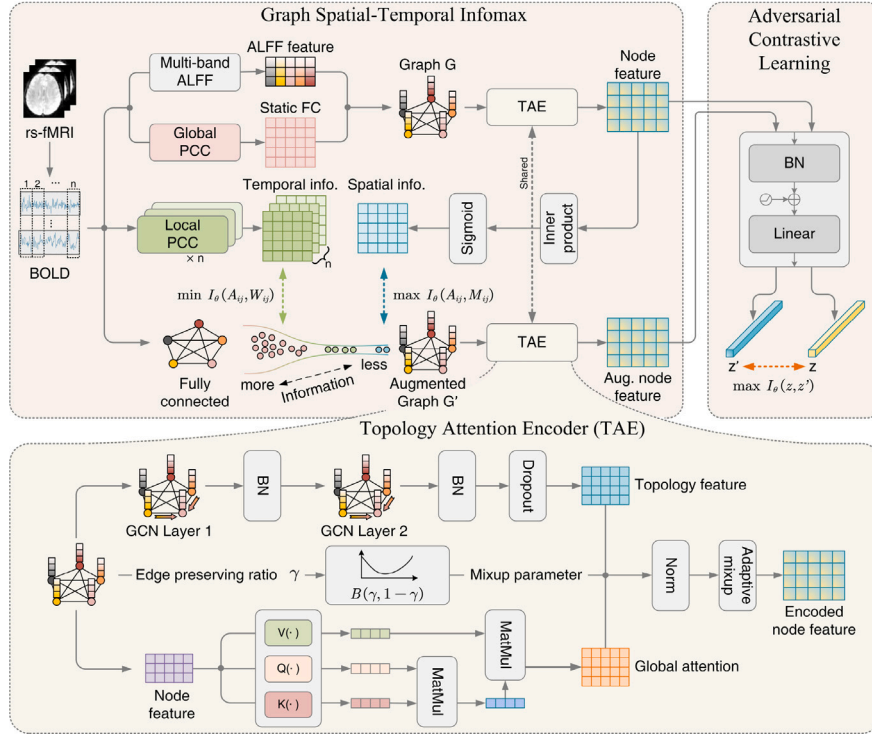


Fig. 2. The framework of the proposed GraSTI-ACL. We first construct the original graph G using rs-fMRI BOLD time series and multi-band ALFF features. During this process, we update the adjacency matrix of the original graph by considering both global spatial information and local temporal dynamics of the nodes. This step ensures a tighter coupling between the adjacency matrix and node information. The original graph G and augmented graph G' are inputted into an encoder (TAE) to update the node features. Through a projection head, we obtain the graph representations z and z' for final optimization.

Combining (6) and (7), the information lower bound to be maximized can be described as:

$$IB = \int p(A_{ij}, M_{ij}) \log q(M_{ij} | A_{ij}) dA_{ij} dM_{ij} - \beta \int p(A_{ij}, W_{ij}) \log \frac{p(A_{ij} | W_{ij})}{r(A_{ij})} dA_{ij} dW_{ij}. \quad (8)$$

However, optimizing (8) is challenging, thus we estimate the empirical distribution:

$$p(M_{ij}, W_{ij}) = \frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N \delta_{M_{ij}}(M) \delta_{W_{ij}}(W), \quad (9)$$

where $\delta(\cdot)$ denotes the Dirac delta function, which represents the empirical distribution as a sum of point masses at the training samples. Eq. (8) then becomes:

$$IB = \frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N \int p(A | M_{ij}) \log q(M_{ij} | A) - \beta p(A | W_{ij}) \log \frac{p(A | W_{ij})}{r(A)} dA. \quad (10)$$

By using reparameterization trick, we rewrite $\int p(A | M_{ij}) dA$ as $\int p(\epsilon) d\epsilon$ to facilitate gradient updates, where $\epsilon \sim \mathcal{N}(0, I)$. Eq. (10) then becomes the final objective function:

$$IB = \frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N \left[\int p(\epsilon_1) \log q(M_{ij} | f(W_{ij}, \epsilon_1)) - \beta p(A | W_{ij}) \log \frac{p(A | W_{ij})}{r(A)} dA \right], \quad (11)$$

where N is the number of ROI, $\epsilon_1 \sim p(\epsilon) = \mathcal{N}(0, I)$ is the Gaussian random variable. The detailed proofs of (5)–(11) are given in the Appendix A and interested readers may refer to it for further details.

Noting that the latter term is the Kullback–Leibler (KL) divergence between p and r , and since the distribution q of the learned adjacency matrix is ideally an independent Bernoulli distribution, the first term becomes a binary cross entropy. Also, it making gradient updates challenging. We address this issue via following reparameterization trick as:

$$A = \sigma \left(\frac{1}{\tau} \left(\log \frac{\epsilon_2}{1 - \epsilon_2} + \log \frac{W_{ij}}{1 - W_{ij}} \right) \right), \quad (12)$$

where $\epsilon_2 \sim U(0, 1)$ is the uniform distribution, σ is the sigmoid function, and τ denotes the temperature parameter.

Rewriting the edge weight matrix $W \in \mathbb{R}^{N \times N}$ as a dynamic edge weight matrix $\mathcal{W} \in \mathbb{R}^{T \times N \times N}$, where T represents the number of segments obtained by dividing the continuous averaged BOLD signal time series, we obtain the final node information bottleneck:

$$\hat{I}_N = \frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N [L_{CE}(A_{ij}, M_{ij}) - \beta D_{KL}(p(A | \mathcal{W}) || r(A_{ij}))]. \quad (13)$$

where $L_{CE}(A_{ij}, M_{ij})$ denotes the cross entropy loss between A_{ij} and $M_{i,j}$, $D_{KL}(\cdot || \cdot)$ is the operation of calculating the KL divergence.

Intuitively, maximizing the first term makes the optimized adjacency matrix more closely resemble the nodal feature vector dot product, while minimizing the second term brings the adjacency matrix distribution closer to a Gaussian distribution. To find a more accurate information bottleneck that balances information sufficiency and necessity in the adjacency matrix, we write $\hat{I}_N$ as:

$$I_N = \frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N [L_{CE}(A_{ij}, M_{ij}) + \beta D_{KL}(p(A | \mathcal{W}) || r(A_{ij}))]. \quad (14)$$

We further adopt adversarial optimization, thus our final node IB optimization function is:

$$\min_{\Phi} \max_{\Theta} I_N, \quad (15)$$

where Φ are the network parameters encoding the adjacency matrix A , and Θ are the network parameters encoding the node vectors.

After passing through node IB, we extract the necessary spatial and temporal information from the nodes while discarding redundant data. It is important to note that this process fundamentally alters the original graph. To prevent excessive perturbations that could compromise useful information, we regulate the temperature coefficient in (12) to transform the originally hard-binarized Bernoulli distribution into a soft-binarized one, resulting in the soft sparsification of the original graph's adjacency matrix. In other words, instead of discarding certain edges, we weaken their connections. Such a soft edge dropout enables the numerical values of the initial FC matrix input to be dynamically updated during training, similar to the approach in [Thapaliya et al. \(2025b\)](#), ensuring that we are not constrained by the manually extracted FC matrix. Next, we input the augmented graph into an encoder for graph representation.

3.3. Topology attention encoder

Next, we propose a novel encoder to better update node features in the graph without a strict all-or-nothing edge dropout approach. Instead, we implements soft dropout by setting the temperature coefficient τ in (12) to 1, which aim to prevent overly aggressive perturbations of graph structure. In the meantime, nodes can still aggregate information from neighboring nodes via graph convolution even if their connections are weakened. To adapt to the soft dropout mechanism and expand the receptive field of nodes, we designed a novel encoder. It aims to expand the receptive field of nodes while preserving the feature of graph structure, enabling the capture of information from distant nodes. Furthermore, as the connectivity of the graph diminishes during training, this encoder can appropriately reduce the influence of topological information and focus more on global information.

To balance attention information from a larger receptive field and topology information from neighbors based on connection strength, we draw inspiration from Graphormer ([Ying et al., 2021](#)) and apply the Transformer principle to compute attention scores for all nodes in the graph. These scores serve as weights for message aggregation. The process is as follows:

$$\begin{aligned} W_{\text{Atte}} &= \text{Att}(X_i, X_j), \quad \text{for } i, j = 1 \dots N, \\ X_{\text{Atte}} &= \text{Agg}(X_i, X_j, w_{ij}), \quad \text{for } i, j = 1 \dots N, \end{aligned} \quad (16)$$

where $\text{Att}(\cdot)$ calculates the attention score between the i th and j th node features, resulting in the attention score matrix W_{Atte} with elements w_{ij} . $\text{Agg}(\cdot)$ aggregates messages based on the attention score matrix, updating the node features X to X_{Atte} , which includes global information.

This approach addresses the issue of GCN's limited receptive field but loses some topology information. Therefore, we also use a GCN to aggregate the topology information:

$$\begin{aligned} X_{\text{Topo}} &= \text{GCN}(X, A) \\ \text{GCN}(X, A) &= \sigma \left(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} X \Theta \right), \end{aligned} \quad (17)$$

where $\tilde{A} = A + I$ is the adjacency matrix with added self-loops, $\tilde{D}$ is the degree matrix of $\tilde{A}$, with $\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}$, Θ is the parameter of network, and σ is the activation function, such as ReLU. The updated node features X are denoted as X_{Topo} , containing topology information.

It is important to note that X_{Atte} and X_{Topo} are independently obtained based on the original X and have the same shape. We now need to mix them according to a certain weighting. We use the Mixup ([Zhang](#)

[et al., 2017](#)) operation, where the weighting is calculated using the Beta function:

$$\lambda = B(x, y) = \int_0^1 t^{x-1} (1-t)^{y-1} dt. \quad (18)$$

Unlike the usual parameters $x = y = 0.4$ used in Mixup, which bias the weight towards two extremes, we use $x = \gamma$ and $y = 1 - \gamma$, where $\lambda = B(\gamma, 1 - \gamma)$. Here, γ represents the strength of retained edges in the graph, calculated as the average of all elements in the adjacency matrix. The final representation of global and structural message passing is:

$$X_{\text{Update}} = X_{\text{Topo}} + \lambda X_{\text{Atte}}. \quad (19)$$

In this way, the final node representation X_{Update} adaptively contains different proportions of global and structural messages based on the degree of edge retention. The use of the Beta function introduces a degree of randomness, ensuring the effectiveness of the final graph representation.

3.4. Adversarial contrastive learning

Now, having obtained the original and augmented graphs through the node IB and updated node features via the improved encoder TAE, suitable graph representations can be learned for downstream tasks. We project both the original and augmented graphs to lower-dimensional representations $z \in \mathbb{R}^D$, where D is the dimension of the graph representation, using a shared projection head. To balance robustness and effectiveness, we input graphs in batches, treating the representation of each graph and its augmented version as positive samples and all other augmented graph representations in the batch as negative samples. Such a process can be referred to graph representation IB. By maximizing the mutual information between positive pairs and minimizing it between negative pairs, our representation IB optimization objective I_R becomes:

$$\begin{aligned} I_R &= I(f_{\Theta}(G), f_{\Theta}(t_{\Phi}(G))) \\ \rightarrow \hat{I}_R &= \frac{1}{m} \sum_{i=1}^m \log \frac{\exp(\text{sim}(z_{i,1}, z_{i,2}))}{\sum_{i'=1, i' \neq i}^m \exp(\text{sim}(z_{i,1}, z_{i',2}))}, \end{aligned} \quad (20)$$

where m is the number of graphs in the batch, and $\text{sim}(\cdot)$ denotes the cosine similarity function. Similarly, through adversarial optimization of (20), we have:

$$\min_{\Psi} \max_{\Omega} I_R, \quad (21)$$

where Ω represents the learnable parameters of the projection head, and Ψ represents the learnable parameters of the TAE.

In this process, we ensure the effectiveness of graph representations, where adversarial optimization prevents catastrophic effects from aggressive perturbations, balancing effectiveness and robustness. Put everything together, the final optimization function is:

$$\min_{\Phi, \Psi} \max_{\Theta, \Omega} (I_R + I_N). \quad (22)$$

4. Experiments and results

4.1. Datasets and preprocessing

We select three brain disease datasets for method validation in this study: (1) An AD dataset from Alzheimer's Disease Neuroimaging Initiative (ADNI) ([Petersen et al., 2010](#)). It is a comprehensive longitudinal multicenter study established to develop clinical, imaging, genetic, and biochemical biomarkers for the early detection and monitoring of Alzheimer's disease. (2) An MDD dataset from Strategic Research Program for Brain Sciences (SRPBS) ([Tanaka et al., 2021](#)). And (3) A self-collected BD dataset by our research team. The data acquisition procedures of this experiment were approved by the Human

Table 1
Demographic information of three datasets used in our experiments.

Dataset	F/M	Patient/Normal	Age(mean $\pm$ std)	Total
AD (Petersen et al., 2010)	68/55	61/32	70 $\pm$ 5	123
MDD (Tanaka et al., 2021)	277/209	247/239	40 $\pm$ 14	486
BD	60/58	65/64	31 $\pm$ 8	118

Research Ethics Committee of Shenzhen Mental Health Center, and all participants in this experiment signed an informed consent form. Demographic information of these three datasets can be found in Table 1.

The preprocessing was conducted using the Data Processing Assistant for Resting-State fMRI (DPARSF) (Yan and Zang, 2010). The preprocessing steps involved file format conversion, time point trimming, motion correction, normalization, Gaussian smoothing, detrending, denoising, and covariate removal. In this paper, we use the Automated Anatomical Labeling (AAL) atlas for BOLD signal extraction, thus $N = 90$ represents the number of ROIs in AAL by excluding the cerebellum. Each node feature consists of ALFF features from three different frequencies: slow-5 0.01 Hz–0.027 Hz, slow-4 0.027 Hz–0.073 Hz, and classical 0.01 Hz–0.08 Hz, which leads to $d = 3$.

4.2. Baselines and metrics

The baseline models employed in this study include GCN (Kipf and Welling, 2016), GAT (Veličković et al., 2017), BrainGNN (Li et al., 2021), HiGCN (Jiang et al., 2020), AutoGCL (Yin et al., 2022), ContrastPool (Xu et al., 2024), AD-GCL (Suresh et al., 2021), AGCL (Zhang et al., 2023), DSAM (Thapaliya et al., 2025b), BrainRGIN (Thapaliya et al., 2025a), BNT (Kan et al., 2022), STAGIN (Kim et al., 2021), and STIGR (Liu et al., 2025). We evaluated our proposed method against these baselines on the above mentioned three datasets. The classification performance metrics used were accuracy, F1 score, sensitivity, specificity, precision and area under the curve (AUC), with results presented as mean $\pm$ standard deviation. Our method was tested using five-fold cross-validation, with the learning rate set to 0.0005. After obtaining the learned graph-level representation z , we employ a Linear Support Vector Classification (LinearSVC) for classification. The hyperparameters are selected through a 5-fold cross-validated grid search. Prior to training, the embeddings are standardized using a StandardScaler. Model selection is performed over a range of regularization strengths ($C \in 0.001, 0.01, 0.1, 1, 10, 100, 1000$). Baseline models followed the parameter settings reported in their original studies. For models without specific input requirements, we used the same ALFF features as our method for node representation. For models that explicitly require BOLD time series or FC matrices (e.g., HiGCN, ContrastPool, DSAM, BrainRGIN, BNT, STAGIN, STIGR), we adhered to the original input settings.

We conducted our experiments on an NVIDIA RTX 3090 GPU with 24 GB of VRAM. For parameter count, the main model contains about 422k parameters, the auxiliary view module about 420k parameters, with approximately 415k parameters shared between them.

4.3. Comparison results

The results of baseline comparative experiment are shown in Tables 2–4. The values are expressed as percentages, with the mean and standard deviation for each metric. The best performance in each metric highlighted in bold and the second-best performance underlined. From the results, we can see that our method shows superior performance on all datasets. Specifically, On AD, GraSTI-ACL achieves the highest accuracy (78.13%) and sensitivity (86.56%), with F1 score and AUC ranking second-best. For MDD, it leads in accuracy (62.56%) and F1 score (62.51%), and a second-best precision (63.69%). On BD, it outperforms all methods in every metric except F1, achieving the

highest accuracy (72.86%), sensitivity (72.33%), specificity (78.26%), precision (76.74%), and AUC (73.73%), with a competitive F1 score of 69.42%.

It should be noted that, compared to basic baselines like GCN and GAT, our method improves accuracy significantly on all three datasets. Against dynamic graph construction methods such as STAGIN and STIGR, our method consistently outperforms them in accuracy across all datasets. Although STIGR shows competitive results on the MDD dataset, achieving 61.33% accuracy compared to our 62.56% and outperforms our method in specificity, precision, and AUC, it lags significantly behind in F1 score and sensitivity, both falling below 50%. Also, our model exhibits more stable and generalized performance across datasets. This highlights the robustness and adaptability of our approach, particularly due to the effectiveness of the TAE module in capturing discriminative temporal features. When comparing to the state-of-the-art DSAM, which similarly does not construct dynamic graphs, our method consistently achieves higher accuracy across all three datasets. Specifically, on the AD dataset, our model outperforms DSAM by 17.3% (78.13% vs. 60.83%), while on the MDD dataset, the improvement reaches 5.04% (62.56% vs. 57.52%), and on the BD dataset, the gain is 24.35% (72.68% vs. 48.33%). This consistent advantage can be attributed to our incorporation of the information bottleneck principle, which facilitates the selective filtering of temporal dynamics, leading to more informative temporal representations. Across other models, accuracy gains range from 0.13% to 16.59% (AD), 2.06% to 10.92% (MDD), and 2.53% to 19.68% (BD). Overall, the results demonstrate that the proposed GraSTI-ACL method shows superior performance across a variety of metrics, making it a promising method for brain disorder-related tasks.

4.4. Ablations

To investigate the effectiveness of different components of our model, we conducted ablation experiments to evaluate the impact of key components in GraSTI-ACL, including the TAE and two optimization terms: cross-entropy loss L_{CE} and KL divergence loss L_{KL} . The study is conducted on three datasets same as previous baseline comparison. The results are shown in Table 5. In the table, each row corresponds to a different combination of components, showing their effect on model performance in terms of accuracy (mean $\pm$ standard deviation). The best performance in each dataset is highlighted in bold, while the second-best is underlined.

The results show that TAE alone provides a strong baseline. However, in the AD dataset, the combination of TAE and L_{CE} (Row 4) achieves the highest accuracy (79.63%), outperforming other configurations. This can be attributed to the spatially localized nature of AD-related disruptions, such as in the hippocampus and precuneus (Cavanna and Trimble, 2006; Rao et al., 2022), which are well-captured by the TAE’s expanded receptive field. The L_{CE} loss further reinforces node-level alignment by maximizing mutual information between the learned adjacency structure and nodal interactions. Together, they enhance the model’s ability to extract and emphasize discriminative spatial features critical for AD diagnosis.

When L_{KL} is used without L_{CE} (Row 3), performance gains are limited, likely due to over-smoothing from KL regularization. Without L_{CE} to guide structure learning, the adjacency matrix becomes overly uniform, weakening the model’s ability to capture key localized connectivity patterns critical for brain disorder classification. This suggests that entropy guidance strongly benefits from temporally informed representations in this context. Distribution-level regularization is most effective when combined with meaningful temporal features, as TAE enhances the model’s ability to structure representations in a biologically relevant way. However, TAE without proper optimization signals (i.e., L_{CE} or L_{KL}) may lead to overfitting or unstable convergence in certain datasets.

Table 2

Performance comparison on the AD dataset.

Method	Accuracy (%)	F1 score (%)	Sensitivity (%)	Specificity (%)	Precision (%)	AUC (%)
GCN (Kipf and Welling, 2016)	54.57 ± 4.93	51.48 ± 9.80	50.30 ± 16.87	65.82 ± 11.54	59.03 ± 13.37	59.86 ± 8.05
GAT (Veličković et al., 2017)	59.20 ± 8.84	52.93 ± 11.92	48.09 ± 14.81	72.97 ± 9.73	63.53 ± 17.42	61.53 ± 0.92
BrainGNN (Li et al., 2021)	64.32 ± 12.52	84.61 ± 13.88	71.94 ± 10.97	74.06 ± 11.14	72.76 ± 10.50	70.28 ± 8.52
HiGCN (Jiang et al., 2020)	61.54 ± 17.42	58.21 ± 19.91	63.66 ± 21.20	70.87 ± 25.58	66.90 ± 31.57	57.43 ± 18.35
AutoGCL (Yin et al., 2022)	64.94 ± 8.77	60.09 ± 23.31	63.39 ± 29.66	70.00 ± 48.30	68.86 ± 23.36	64.40 ± 11.23
ContrastPool (Xu et al., 2024)	63.53 ± 12.39	54.96 ± 28.25	57.56 ± 33.09	68.98 ± 35.47	58.00 ± 32.46	63.27 ± 13.26
AD-GCL (Suresh et al., 2021)	76.43 ± 4.61	76.00 ± 5.03	76.67 ± 14.00	77.26 ± 8.95	77.83 ± 8.41	76.55 ± 3.96
AGCL (Zhang et al., 2023)	77.23 ± 1.97	75.29 ± 3.30	78.16 ± 5.17	82.81 ± 6.69	81.31 ± 5.25	76.69 ± 1.40
DSAM (Thapaliya et al., 2025b)	60.83 ± 5.54	54.81 ± 7.86	79.85 ± 33.32	42.56 ± 34.51	73.27 ± 8.94	64.24 ± 5.70
BrainRGIN (Thapaliya et al., 2025a)	68.37 ± 8.37	69.73 ± 7.15	74.31 ± 19.23	62.63 ± 36.89	70.75 ± 17.47	68.46 ± 11.84
BNT (Kan et al., 2022)	78.00 ± 8.14	77.46 ± 8.06	83.67 ± 15.02	71.89 ± 11.76	79.67 ± 10.06	88.27 ± 7.12
STAGIN (Kim et al., 2021)	66.57 ± 24.32	62.34 ± 25.49	56.03 ± 23.65	77.18 ± 28.33	68.00 ± 27.07	72.67 ± 28.77
STIGR (Liu et al., 2025)	69.30 ± 12.30	57.18 ± 29.70	56.03 ± 24.81	79.74 ± 25.52	67.25 ± 37.29	74.79 ± 18.58
GraSTI-ACL	78.13 ± 7.79	78.81 ± 5.39	86.56 ± 9.53	77.97 ± 11.98	78.72 ± 10.23	79.10 ± 6.94

Table 3

Performance comparison on the MDD dataset.

Method	Accuracy (%)	F1 score (%)	Sensitivity (%)	Specificity (%)	Precision (%)	AUC (%)
GCN (Kipf and Welling, 2016)	52.05 ± 4.37	45.54 ± 7.10	40.17 ± 8.21	64.89 ± 10.10	54.56 ± 9.07	52.53 ± 4.74
GAT (Veličković et al., 2017)	48.75 ± 3.12	49.66 ± 4.44	50.56 ± 8.43	43.76 ± 8.80	49.82 ± 3.36	48.78 ± 5.32
BrainGNN (Li et al., 2021)	60.50 ± 2.93	58.28 ± 3.45	68.12 ± 21.23	51.27 ± 19.67	61.63 ± 3.32	59.36 ± 3.40
HiGCN (Jiang et al., 2020)	59.06 ± 3.30	54.41 ± 3.62	49.82 ± 5.17	68.07 ± 7.34	60.53 ± 5.19	58.15 ± 22.71
AutoGCL (Yin et al., 2022)	52.90 ± 5.67	50.37 ± 9.00	50.23 ± 18.51	57.09 ± 21.52	57.88 ± 13.10	53.47 ± 4.98
ContrastPool (Xu et al., 2024)	51.82 ± 5.38	40.02 ± 2.68	46.48 ± 38.47	56.70 ± 34.47	49.93 ± 24.18	51.59 ± 5.23
AD-GCL (Suresh et al., 2021)	58.22 ± 4.29	61.87 ± 4.06	69.93 ± 8.79	54.41 ± 9.67	58.17 ± 6.72	60.08 ± 3.61
AGCL (Zhang et al., 2023)	59.65 ± 3.40	62.15 ± 3.16	67.61 ± 6.72	53.55 ± 6.06	60.14 ± 9.43	59.15 ± 1.62
DSAM (Thapaliya et al., 2025b)	57.52 ± 7.87	54.56 ± 7.44	75.89 ± 27.66	39.66 ± 17.30	62.83 ± 12.10	61.13 ± 9.16
BrainRGIN (Thapaliya et al., 2025a)	51.64 ± 2.65	47.54 ± 25.60	56.92 ± 34.75	48.50 ± 31.89	42.25 ± 23.72	52.71 ± 3.16
BNT (Kan et al., 2022)	53.31 ± 10.17	40.33 ± 16.85	<u>75.56 ± 43.32</u>	32.56 ± 46.48	33.70 ± 20.77	72.84 ± 9.24
STAGIN (Kim et al., 2021)	54.72 ± 4.16	61.84 ± 0.75	72.11 ± 5.81	36.88 ± 14.07	54.64 ± 3.87	54.36 ± 5.45
STIGR (Liu et al., 2025)	61.33 ± 4.26	49.15 ± 15.21	41.94 ± 16.53	78.57 ± 11.79	67.34 ± 6.00	73.21 ± 2.92
GraSTI-ACL	62.56 ± 6.95	62.51 ± 8.31	64.88 ± 8.94	63.17 ± 10.51	<u>63.69 ± 11.07</u>	61.00 ± 3.57

Table 4

Performance comparison on the BD dataset.

Method	Accuracy (%)	F1 score (%)	Sensitivity (%)	Specificity (%)	Precision (%)	AUC (%)
GCN (Kipf and Welling, 2016)	58.48 ± 4.77	56.13 ± 6.94	61.19 ± 12.27	59.74 ± 9.26	55.68 ± 14.31	59.67 ± 5.38
GAT (Veličković et al., 2017)	57.64 ± 4.34	55.78 ± 9.24	62.26 ± 13.65	57.13 ± 11.55	54.47 ± 1.56	55.66 ± 13.49
BrainGNN (Li et al., 2021)	70.33 ± 7.17	68.81 ± 6.66	61.38 ± 9.30	76.91 ± 9.12	69.51 ± 6.34	69.39 ± 10.95
HiGCN (Jiang et al., 2020)	65.98 ± 26.82	60.31 ± 20.69	62.57 ± 26.82	69.05 ± 19.4	65.51 ± 19.6	64.86 ± 17.01
AutoGCL (Yin et al., 2022)	53.18 ± 12.89	46.32 ± 18.60	50.75 ± 24.60	58.69 ± 27.11	53.95 ± 25.07	54.72 ± 9.58
ContrastPool (Xu et al., 2024)	53.71 ± 11.75	44.10 ± 23.14	50.67 ± 33.09	59.52 ± 30.81	43.00 ± 24.7	55.10 ± 10.57
AD-GCL (Suresh et al., 2021)	67.79 ± 11.72	64.39 ± 7.11	65.76 ± 10.12	<u>78.00 ± 11.52</u>	<u>70.00 ± 17.08</u>	70.32 ± 12.31
AGCL (Zhang et al., 2023)	70.29 ± 8.31	66.14 ± 9.18	66.56 ± 15.26	77.79 ± 14.84	68.55 ± 16.22	71.28 ± 7.97
DSAM (Thapaliya et al., 2025b)	48.33 ± 14.23	40.02 ± 17.74	40.00 ± 44.87	63.08 ± 37.53	37.18 ± 20.94	51.68 ± 13.20
BrainRGIN (Thapaliya et al., 2025a)	58.40 ± 13.44	56.48 ± 18.01	66.33 ± 32.73	49.01 ± 35.30	55.91 ± 14.24	57.67 ± 9.75
BNT (Kan et al., 2022)	64.35 ± 11.40	62.32 ± 9.82	58.06 ± 18.37	67.19 ± 23.48	67.03 ± 14.64	70.94 ± 9.29
STAGIN (Kim et al., 2021)	55.04 ± 4.72	42.11 ± 18.61	44.00 ± 28.67	63.85 ± 20.87	50.20 ± 3.20	59.79 ± 9.07
STIGR (Liu et al., 2025)	54.24 ± 1.38	70.32 ± 1.16	56.92 ± 36.60	36.73 ± 34.79	54.24 ± 1.38	61.86 ± 8.73
GraSTI-ACL	72.86 ± 10.17	<u>69.42 ± 7.83</u>	72.33 ± 17.80	78.26 ± 14.92	76.74 ± 15.83	73.73 ± 5.32

Table 5Ablation study results. The impact of TAE, L_{CE} and L_{KL} on AD, MDD and BD diagnostic accuracy (%) is evaluated.

	TAE	L_{CE}	L_{KL}	AD	MDD	BD
1	✓			76.43 ± 10.23	61.31 ± 1.98	71.23 ± 5.34
2		✓		77.33 ± 5.75	58.43 ± 3.12	69.53 ± 6.49
3			✓	77.20 ± 8.98	59.66 ± 3.85	71.23 ± 6.51
4	✓	✓		79.63 ± 5.81	61.51 ± 3.16	72.84 ± 7.23
5	✓		✓	77.26 ± 9.51	60.70 ± 7.67	71.23 ± 5.34
6		✓	✓	77.33 ± 6.77	59.68 ± 4.78	70.36 ± 5.77
7	✓	✓	✓	<u>78.13 ± 7.79</u>	62.56 ± 6.95	72.86 ± 10.17

The combination of L_{CE} and L_{KL} without TAE (Row 6) does not provide a significant advantage, suggesting that TAE is essential for effectively integrating global information into GCNs. The full model (Row 7) achieves the best overall performance, excelling in MDD (62.56%) and BD (72.86%), while maintaining strong AD diagnostic

accuracy (second-best 78.13%). These results confirm that all three components contribute together to improving performance, with TAE playing a central role in feature integration, and the loss terms refining the optimization process.

Furthermore, ablation studies results of dynamic time information are shown in Table 6. It presents the results of an ablation study examining the impact of dynamic time information on model performance across three datasets: AD, MDD and BD. The comparison is made between two settings: static time information and dynamic time information. The results indicate that dynamic time information generally improves classification performance, particularly in accuracy, F1 score and AUC. In the AD dataset, using dynamic time information leads to a significant increase in sensitivity (86.56% vs. 81.32%) and AUC (79.10% vs. 78.76%), though static time information performs slightly better in specificity and precision. The most notable improvement is observed in the MDD dataset, where dynamic time information results in a substantial increase in accuracy (62.56% vs. 58.64%),

Table 6
Ablation studies results of dynamic time information.

Dataset	TimeInfo	Accuracy	F1 score	Sensitivity	Specificity	Precision	AUC
AD	Static	77.93 $\pm$ 6.97	78.07 $\pm$ 4.80	81.32 $\pm$ 9.64	80.78 $\pm$ 15.31	80.88 $\pm$ 11.70	78.76 $\pm$ 7.54
	Dynamic	78.13 $\pm$ 7.79	78.81 $\pm$ 5.39	86.56 $\pm$ 9.53	77.97 $\pm$ 11.98	78.72 $\pm$ 10.23	79.10 $\pm$ 6.94
MDD	Static	58.64 $\pm$ 2.97	62.38 $\pm$ 2.54	67.54 $\pm$ 1.44	57.37 $\pm$ 13.51	59.33 $\pm$ 10.89	60.78 $\pm$ 4.92
	Dynamic	62.56 $\pm$ 6.95	62.51 $\pm$ 8.31	64.88 $\pm$ 8.94	63.17 $\pm$ 10.51	63.69 $\pm$ 11.07	61.00 $\pm$ 3.57
BD	Static	72.79 $\pm$ 10.99	69.11 $\pm$ 11.31	73.15 $\pm$ 9.46	77.01 $\pm$ 13.97	71.29 $\pm$ 16.85	70.87 $\pm$ 10.51
	Dynamic	72.86 $\pm$ 10.17	69.42 $\pm$ 7.83	72.33 $\pm$ 17.80	78.26 $\pm$ 14.92	76.74 $\pm$ 15.83	73.73 $\pm$ 5.32

Table 7
The experiment results between the original AGCL and modified AGCL-TAE on 3 datasets. All metrics are shown as percentage (%).

Dataset	Method	Accuracy	F1 score	Sensitivity	Specificity	Precision	AUC
AD	AGCL	77.23 $\pm$ 1.97	75.29 $\pm$ 3.30	78.16 $\pm$ 5.17	82.81 $\pm$ 6.69	81.31 $\pm$ 5.25	76.69 $\pm$ 1.40
	AGCL-TAE	78.03 $\pm$ 4.15	77.96 $\pm$ 3.35	78.55 $\pm$ 12.39	79.13 $\pm$ 11.86	80.12 $\pm$ 9.25	78.84 $\pm$ 3.40
MDD	AGCL	59.65 $\pm$ 3.40	62.15 $\pm$ 3.16	67.61 $\pm$ 6.72	53.55 $\pm$ 6.06	60.14 $\pm$ 9.43	59.15 $\pm$ 1.62
	AGCL-TAE	61.32 $\pm$ 2.95	63.30 $\pm$ 4.71	67.85 $\pm$ 5.38	56.12 $\pm$ 3.98	60.84 $\pm$ 4.05	61.18 $\pm$ 2.70
BD	AGCL	70.29 $\pm$ 8.31	66.14 $\pm$ 9.18	66.56 $\pm$ 15.26	77.79 $\pm$ 14.84	68.55 $\pm$ 16.22	71.28 $\pm$ 7.97
	AGCL-TAE	71.20 $\pm$ 3.01	66.29 $\pm$ 5.51	69.06 $\pm$ 6.89	80.39 $\pm$ 10.77	73.33 $\pm$ 10.93	71.60 $\pm$ 7.37

Table 8
The experiment results between the original AD-GCL and modified AD-GCL-TAE on 3 datasets. All metrics are shown as percentage (%).

Dataset	Method	Accuracy	F1 score	Sensitivity	Specificity	Precision	AUC
AD	AD-GCL	76.43 $\pm$ 4.61	76.00 $\pm$ 5.03	76.67 $\pm$ 14.00	77.26 $\pm$ 8.95	77.83 $\pm$ 8.41	76.55 $\pm$ 3.96
	AD-GCL-TAE	77.23 $\pm$ 1.97	76.58 $\pm$ 3.51	79.01 $\pm$ 15.88	78.69 $\pm$ 7.15	80.50 $\pm$ 10.97	77.21 $\pm$ 6.19
MDD	AD-GCL	58.22 $\pm$ 4.29	61.87 $\pm$ 4.06	69.93 $\pm$ 8.79	54.41 $\pm$ 9.67	58.17 $\pm$ 6.72	60.08 $\pm$ 3.61
	AD-GCL-TAE	62.55 $\pm$ 3.85	63.44 $\pm$ 4.61	68.05 $\pm$ 12.52	61.08 $\pm$ 6.58	63.19 $\pm$ 6.47	62.97 $\pm$ 3.53
BD	AD-GCL	67.79 $\pm$ 11.72	64.39 $\pm$ 7.11	65.76 $\pm$ 10.12	78.00 $\pm$ 11.52	70.00 $\pm$ 17.08	70.32 $\pm$ 12.31
	AD-GCL-TAE	69.53 $\pm$ 8.98	66.48 $\pm$ 8.49	67.79 $\pm$ 11.22	80.19 $\pm$ 12.04	70.83 $\pm$ 13.44	71.59 $\pm$ 7.37

specificity (63.17% vs. 57.37%), and precision (63.69% vs. 59.33%), demonstrating its effectiveness in this scenario. For the BD dataset, dynamic time information shows slightly higher accuracy (72.86% vs. 72.79%), F1 score (69.42% vs. 69.11%), and AUC (73.73% vs. 70.87%), while maintaining balanced Sensitivity and Specificity. Overall, these findings suggest that incorporating dynamic time information enhances classification stability and generalizability, particularly in MDD and BD datasets, while showing a trade-off in certain metrics for AD.

4.5. Effectiveness of TAE

Here we show the effectiveness of our designed TAE by applying it to A-GCL and AD-GCL, which use the same graph augmentation strategy. The results are shown in [Tables 7 and 8](#). [Table 7](#) presents a comparative analysis of two methods, original AGCL and its modified version AGCL-TAE, evaluated on three different datasets: AD, MDD, and BD. The performance metrics include accuracy, F1 score, sensitivity, specificity, precision, and AUC, all reported as percentages with mean $\pm$ standard deviation. Across all datasets, AGCL-TAE consistently outperforms the original AGCL in most metrics, demonstrating its effectiveness in rs-fMRI-based brain disorders diagnostic tasks. Notably, in the AD dataset, AGCL-TAE achieves higher accuracy (78.03% vs. 77.23%), F1 score (77.96% vs. 75.29%), and AUC (78.84% vs. 76.69%), indicating better overall classification performance. Similarly, for the MDD and BD datasets, AGCL-TAE improves all metrics, suggesting better classification stability. These results highlight the effectiveness of the proposed TAE improvement in enhancing AGCL's performance across different neurological and psychiatric conditions. Additionally, [Table 8](#) presents a comparison between the original AD-GCL model and its improved version, AD-GCL-TAE. The results indicate that AD-GCL-TAE consistently outperforms AD-GCL across almost all metrics, demonstrating the effectiveness of the proposed TAE improvement. These results further confirms the effectiveness of TAE, leading to

improved classification accuracy and overall model reliability across different neurological and psychiatric conditions.

We also incorporate TAE into LightGCL ([Cai et al., 2023](#)), a model similar to AD-GCL to validate its usefulness via different types of datasets. Following the experimental settings outlined in the ([Cai et al., 2023](#)), we used the same datasets and evaluation metrics for recommendation. The results are presented in [Table 9](#). It compares the performance of the original LightGCL model and its improved version, LightGCL-TAE, across five datasets: Yelp, Gowalla, ML-10M, Amazon, and Tmall. The evaluation metrics include Recall at 20 (R@20), Normalized Discounted Cumulative Gain at 20 (N@20), Recall at 40 (R@40), and NDCG at 40 (N@40), which assess the ranking quality of recommendations. The results indicate that LightGCL-TAE exhibits slight improvements over LightGCL in some datasets, while showing marginal declines in others. Specifically, in the Yelp and Amazon datasets, LightGCL-TAE outperforms LightGCL across all four metrics, demonstrating enhanced recommendation quality. However, in the Gowalla and ML-10M datasets, LightGCL-TAE experiences a small performance drop in all metrics, suggesting that the improvements introduced by TAE may not generalize equally well across different types of datasets. The Tmall dataset shows no significant difference between the two methods, indicating that the enhancement has a minimal impact in this scenario.

Overall, the TAE module is particularly effective for rs-fMRI data, where both local and global temporal dependencies are crucial for modeling brain dynamics. In such settings, integrating temporal signals into the graph helps capture biologically meaningful interactions across spatial and temporal scales. In contrast, non-fMRI datasets like those used in graph recommendation (e.g., LightGCL) often rely more on local structural patterns, and the temporal context is either weak or less informative. This difference partly explains why the performance gain from TAE is less consistent on non-fMRI datasets.

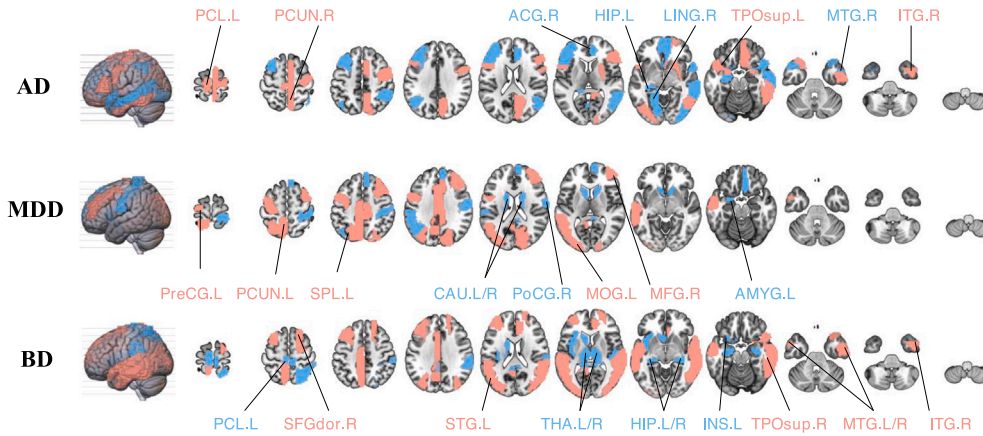


Fig. 3. Identified disease-related brain regions in AD, MDD and BD datasets: red and blue represent regions with increased and decreased connectivity respectively by comparing subjects with diseases to HC.

Table 9

The results of original LightGCL and modified LightGCL-TAE on 5 datasets.

Dataset	Method	R@20	N@20	R@40	N@40
Yelp	LightGCL	0.0793	0.0668	0.1292	0.0852
	LightGCL-TAE	0.0801	0.0677	0.1302	0.0862
Gowalla	LightGCL	0.1578	0.0935	0.2245	0.1108
	LightGCL-TAE	0.1510	0.0897	0.2171	0.1069
ML-10M	LightGCL	0.2613	0.3106	0.3799	0.3387
	LightGCL-TAE	0.2587	0.3076	0.3763	0.3356
Amazon	LightGCL	0.0585	0.0436	0.0933	0.0551
	LightGCL-TAE	0.0598	0.0447	0.0951	0.0563
Tmall	LightGCL	0.0528	0.0361	0.0852	0.0473
	LightGCL-TAE	0.0528	0.0361	0.0856	0.0475

4.6. Disease-related ROIs identification

In medical imaging analysis, complex models' interpretability is as important as their classification performance because of the need for trustworthy and clinically meaningful insight. Our goal is to balance accuracy with connectivity-level interpretability essential for medical imaging. Unlike simpler models (Popov et al., 2024), our approach directly visualizes region interactions and promotes information efficiency (Bessadok et al., 2023). Specifically, when the model achieves optimal performance, the type of graph augmentation applied suggests that such augmentation is beneficial for distinguishing between patients and healthy subjects. Since brain disorder patients often show altered FC, we identify disease-related ROIs based on the average number of retained edges. In our analysis, brain regions with retained edges outside $[\mu - \sigma, \mu + \sigma]$ were considered related to the disease, where μ and σ represent the mean and standard deviation of all brain regions, respectively. The visualization of the ROIs is shown in Fig. 3, where red and blue represent regions with increased and decreased connectivity ROIs, respectively, when comparing subjects with disease to HC.

AD (Fig. 3 Row 1): Increased connectivity is observed in regions like the temporal pole: superior temporal gyrus (TPOsup), precuneus (PCUN), and inferior temporal gyrus (ITG). These areas are involved in memory processing, language comprehension, self-referential processing, and episodic memory, suggesting compensatory hyperconnectivity in response to neurodegeneration (Cavanna and Trimble, 2006). Conversely, decreased connectivity is observed in the hippocampus (HIP) and anterior cingulate cortex (ACG), which are crucial for memory formation and executive function in AD (Rao et al., 2022). Meanwhile, the lingual gyrus (LING) and middle temporal gyrus (MTG), involved in visual processing and semantic memory, also show decreased connectivity (Meulenbroek et al., 2010). These findings align with significant

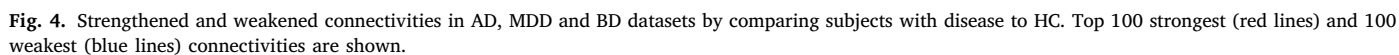
features of AD such as progressive memory impairment and executive dysfunction.

MDD (Fig. 3 Row 2): Increased connectivity is found in the middle occipital gyrus (MOG), precuneus (PCUN), superior parietal lobule (SPL), middle frontal gyrus (MFG) and precentral gyrus (PreCG), which are related to sensory processing and visuospatial attention (Cavanna and Trimble, 2006). This may reflect heightened sensory responsiveness or altered self-referential processing in MDD. Meanwhile, decreased connectivity in the caudate nucleus (CAU), postcentral gyrus (PoCG), and amygdala (AMYG) suggests disruptions in emotional regulation, cognitive control, and reward processing, supporting the hypothesis of fronto-limbic dysfunction in depression (Olson and Berryhill, 2009).

BD (Fig. 3 Row 3): Strengthened connectivity is observed in the middle temporal gyrus (MTG), inferior temporal gyrus (ITG), superior temporal gyrus (STG) and dorsolateral superior frontal gyrus (SFGdor). These regions are linked to language processing, emotional regulation, and motor control, potentially contributing to mood instability and cognitive impairments in BD (Uezato et al., 2009). In contrast, weakened connectivity in the insula (INS), hippocampus (HIP), and thalamus (THA) suggests impaired sensory integration, memory processing, and emotional regulation, which are commonly associated with dysregulated affect and impulsivity in BD (Bechdolf et al., 2012).

In addition to visualizing brain regions with the highest overall connectivity, we aimed to explore the connection strength between brain regions. Fig. 4 shows the strengthened and weakened brain region connections in AD, MDD and BD datasets by comparing subjects with disease and HC. Top 100 strongest (red lines) and 100 weakest (blue lines) connectivities are drawn, with darker shades indicating stronger/weaker connections. The size of the nodes reflects the number of connections a brain region has with other regions. It is important to note that the connectivities presented here reflect pairwise connectivity between specific brain regions, rather than the overall average connectivity across all regions used earlier.

AD is characterized by abnormal disruptions in large-scale brain networks, particularly affecting memory and cognition. Strengthened connectivities between the temporal pole, precuneus, and inferior temporal gyrus suggest compensatory mechanisms against neurodegeneration. These regions are vital for episodic memory and self-referential processing (Li et al., 2024). In contrast, weakened connectivities in the hippocampus, anterior cingulate cortex, middle temporal gyrus, and lingual gyrus, associated with memory consolidation, executive function, and visual processing, align with hippocampal atrophy and network disintegration, contributing to cognitive decline (Sabuncu et al., 2011).



BD has complex connectivity disruptions affecting mood regulation and cognitive control. Strengthened connectivity in the superior, middle, and inferior temporal gyri, temporal pole, and dorsolateral superior frontal gyrus supports language processing, emotional regulation, and executive function (Ha et al., 2009). Increased frontal-temporal connectivity may contribute to mood instability and impaired control. Conversely, weakened connectivities in the insula, hippocampus, thalamus, and paracentral lobule impact salience detection, memory, and sensorimotor integration. Reduced insular and thalamic connectivity suggests disruptions in emotional salience and sensory processing, common in BD (Steullet et al., 2018).

5. Discussions

In this paper, we introduce GraSTI-ACL framework for brain disorders diagnosis based on rs-fMRI. Our framework introduces several novel contributions that distinguish it from existing methods.

Compared to simpler models such as MLP-based approaches that primarily rely on region-averaged features (Popov et al., 2024), our approach explicitly models the network structure and temporal dynamics inherent in functional connectivity data. By representing the brain as a graph and integrating dynamic time-windowed correlations, GraSTI-ACL captures both the spatial configuration and evolving interactions between regions, which are often overlooked in simpler architectures. This design not only enhances interpretability by revealing which connections contribute to classification but also provides a mechanistic perspective on how network disruptions manifest in different disorders. It should be noted that our method is designed to complement, rather than replace, simpler models like MLPs. While MLPs provide strong performance on region-level features, GraSTI-ACL offers an additional layer of interpretability by modeling connectivity patterns and dynamic relationships between regions. This network-level perspective can yield insights that simpler approaches cannot easily capture, making the two approaches mutually informative for understanding brain disorders.

Recent studies have discussed the comparison between generative and contrastive approaches. Approaches such as (Liu et al., 2021; Choi et al., 2023, 2025) and other generative modeling have shown great promise when trained on extensive datasets. While generative self-supervised learning approaches have shown strong potential for learning broad representations from large-scale fMRI data, our method focuses on producing discriminative and interpretable representations specifically tailored for disease classification in limited-data settings. By integrating contrastive learning with an adversarial information bottleneck strategy, our framework selectively captures task-relevant spatial

and temporal patterns while reducing redundant signals. Additionally, modeling brain connectivity explicitly as graphs enhances interpretability by revealing meaningful edge and region-level biomarkers.

5.2. Limitations and future works

This study has several limitations that point to potential directions for future research. From the perspective of input data, the current analysis relies solely on rs-fMRI. This modality may not fully capture the multifaceted nature of brain disorders. Combining structural MRI or diffusion imaging, as well as integrating multi-omics data such as transcriptomics and epigenetics, could offer a more comprehensive understanding of underlying mechanisms.

In terms of methodology, functional connectivity in our current model is estimated using Pearson correlation coefficients. One limitation is the reliance on Pearson correlation to build connectivity graphs, which only captures linear dependencies. Alternative measures such as mutual information or dynamic connectivity metrics may better capture non-linear or time-varying relationships. Future work will explore integrating differentiable mutual information estimators to model nonlinear interactions directly from time series, potentially improving sensitivity to complex brain dynamics.

As discussed above, recent studies have compared generative and contrastive approaches, suggesting that generative methods are more commonly used for building foundational models on large-scale datasets (Liu et al., 2021; Choi et al., 2023, 2025). The ACL module of our model is based on a discriminative framework, while generative strategies could provide deeper insight into the distributional properties of brain networks and be suitable for other scenes. We believe this design complements generative methods, and combining both strategies is a promising direction for future research. In the meantime, clustering-based analysis, as explored in prior works (Thapaliya et al., 2024, 2025a; Li et al., 2021), presents another promising direction to model the modular architecture of the brain. Although these approaches are primarily limited to static or regional information, we believe they provide valuable insights into brain organization. Combining them with our method, which incorporates dynamic connectivity, may offer a more comprehensive and flexible framework for capturing temporal and spatial patterns.

The current validation is limited to publicly available or self-collected datasets with relatively modest sample sizes and controlled acquisition conditions. To more fully evaluate the practical utility and robustness of the method, future work should include prospective clinical studies involving larger, more diverse patient cohorts across multiple sites and scanners. Such studies will be essential to assess the model's generalizability, reproducibility, and potential for integration into real-world clinical workflows.

6. Conclusion

In this paper, we introduce GraSTI-ACL, a novel approach grounded in the information bottleneck principle. GraSTI serves as an efficient graph augmentation method for rs-fMRI time series by constructing an initial graph from fMRI data and applying node-guided edge dropping. By integrating dynamic temporal information into spatial functional brain maps with low computational cost, our method learns robust and effective graph representations through self-supervised adversarial graph contrastive learning, ultimately forming GraSTI-ACL. Experimental results on three brain disorder datasets validate the diagnostic effectiveness of our approach. Furthermore, our model exhibits strong interpretability, as the results across different disease datasets highlight disease-specific brain regions and functional connectivity abnormalities, providing valuable insights into brain disorders.

CRediT authorship contribution statement

Biao He: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Erni Ji:** Writing – review & editing, Writing – original draft, Resources, Data curation, Conceptualization. **Xiaofen Zong:** Writing – review & editing, Writing – original draft, Resources, Data curation, Conceptualization. **Zhen Liang:** Writing – review & editing, Validation, Methodology, Investigation, Funding acquisition, Formal analysis. **Gan Huang:** Writing – review & editing, Validation, Methodology, Investigation, Funding acquisition, Formal analysis. **Li Zhang:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors have no conflict of interest to declare.

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Appendix A. Proof of (5)–(11) from the manuscript

We present our proof process based on the Variational Information Bottleneck (VIB) framework (Alemi et al., 2016). We have a graph $G = (V, E, A, W)$, where $V \in \mathbb{R}^d$ represents the node vectors, with d being the dimensionality of these vectors. $A \in \mathbb{R}^{N \times N}$ is the adjacency matrix to be learned, initialized as a fully connected graph, and $W \in \mathbb{R}^{N \times N}$ is a weight matrix computed based on the node correlations, N is number of nodes. Our objective is to maximize the following function:

$$\max I_{\Theta}(A_{ij}, M_{ij}) - \beta I_{\Theta}(A_{ij}, W_{ij}). \quad (A.1)$$

Here, $I_{\Theta}(A, B)$ denotes the mutual information between A and B , where A_{ij} is the adjacency matrix to be learned, W_{ij} and M_{ij} are the weights and inner products of node vectors between nodes i and j , respectively. In other words, our goal is to maximize the mutual information between the adjacency matrix and the node vectors, while minimizing the mutual information between the adjacency matrix and the weight matrix to prevent it from becoming too similar to the existing weight matrix. β is a hyperparameter.

For the first term, we have:

$$\begin{aligned} I_{\Theta}(A_{ij}, M_{ij}) &= \int p(A_{ij}, M_{ij}) \log \frac{p(A_{ij}, M_{ij})}{p(A_{ij})p(M_{ij})} dA_{ij} dM_{ij} \\ &= \int p(A_{ij}, M_{ij}) \log \frac{p(M_{ij} | A_{ij})}{p(M_{ij})} dA_{ij} dM_{ij}, \end{aligned} \quad (A.2)$$

where $p(A_{ij}, M_{ij})$ is derived using an encoder and a Markov Chain ($M_{ij} \leftrightarrow W_{ij} \leftrightarrow A_{ij}$) as follows:

$$\begin{aligned} p(A_{ij}, M_{ij}) &= \int p(W_{ij}, A_{ij} | M_{ij}) dW_{ij} \\ &= \int p(W_{ij} | M_{ij}) p(A_{ij} | M_{ij}, W_{ij}) dW_{ij} \\ &= \int p(W_{ij} | M_{ij}) p(A_{ij} | W_{ij}) dW_{ij}. \end{aligned} \quad (A.3)$$

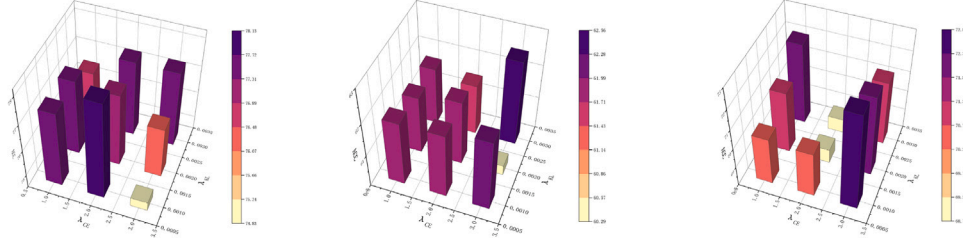


Fig. A.5. The parameter experiment result in AD(left), MDD(middle) and BD(right) dataset, which use accuracy as evaluate metric.

To find the lower bound of (A.2), we use the non-negativity of the KL divergence and assume a variational approximation $q(M_{ij} | A_{ij})$ for $p(M_{ij} | A_{ij})$:

$$\begin{aligned} D_{KL}(p(M_{ij} | A_{ij}) \parallel q(M_{ij} | A_{ij})) &\geq 0 \\ \Rightarrow \int p(M_{ij} | A_{ij}) \log p(M_{ij} | A_{ij}) dA_{ij} & \\ \geq \int p(M_{ij} | A_{ij}) \log q(M_{ij} | A_{ij}) dA_{ij}, \end{aligned} \quad (\text{A.4})$$

substituting into (A.2), we obtain the variational lower bound:

$$\begin{aligned} I_{\Theta}(A_{ij}, M_{ij}) &\geq \int p(A_{ij}, M_{ij}) \log q(M_{ij} | A_{ij}) dA_{ij} dM_{ij} \\ &\quad - \int p(A_{ij}, M_{ij}) \log p(M_{ij}) dA_{ij} dM_{ij}, \end{aligned} \quad (\text{A.5})$$

where

$$\begin{aligned} &\int p(A_{ij}, M_{ij}) \log p(M_{ij}) dA_{ij} dM_{ij} \\ &= \int \log p(M_{ij}) \left(\int p(A_{ij}, M_{ij}) dA_{ij} \right) dM_{ij} \\ &= \int p(M_{ij}) \log p(M_{ij}) dM_{ij} = -H(M_{ij}). \end{aligned} \quad (\text{A.6})$$

This implies that the entropy of the node vector inner products is independent of our optimization objective. Ignoring this term, (A.5) becomes:

$$\begin{aligned} &I_{\Theta}(A_{ij}, M_{ij}) \\ &\geq \int p(A_{ij}, M_{ij}) \log q(M_{ij} | A_{ij}) dA_{ij} dM_{ij}. \end{aligned} \quad (\text{A.7})$$

For the second term in (A.1):

$$\begin{aligned} &I_{\Theta}(A_{ij}, W_{ij}) \\ &= \int p(A_{ij}, W_{ij}) \log \frac{p(A_{ij}, W_{ij})}{p(A_{ij})p(W_{ij})} dA_{ij} dW_{ij} \\ &= \int p(A_{ij}, W_{ij}) \log \frac{p(A_{ij} | W_{ij})}{p(A_{ij})} dA_{ij} dW_{ij} \\ &\quad - \int p(A_{ij}, W_{ij}) \log p(A_{ij}) dA_{ij} dW_{ij}. \end{aligned} \quad (\text{A.8})$$

Again, utilizing the property of KL divergence, with $r(A_{ij})$ as a variational approximation for $p(A_{ij})$, we have:

$$\begin{aligned} &I_{\Theta}(A_{ij}, W_{ij}) \\ &\leq \int p(A_{ij}, W_{ij}) \log \frac{p(A_{ij} | W_{ij})}{r(A_{ij})} dA_{ij} dW_{ij}, \end{aligned} \quad (\text{A.9})$$

combining (A.7) and (A.9), we obtain:

$$\begin{aligned} &I_{\Theta}(A_{ij}, M_{ij}) - \beta I_{\Theta}(A_{ij}, W_{ij}) \\ &\geq \int p(A_{ij}, M_{ij}) \log q(M_{ij} | A_{ij}) dA_{ij} dM_{ij} \\ &\quad - \beta \int p(A_{ij}, W_{ij}) \log \frac{p(A_{ij} | W_{ij})}{r(A_{ij})} dA_{ij} dW_{ij}. \end{aligned} \quad (\text{A.10})$$

Table B.10

Coordinates of identified ROIs in AD dataset.

Type	ROI	x	y	z
strengthen	PCL.R	7.48	-31.59	68.09
	SMA.R	8.62	0.17	61.85
	IFGperc.L	-48.43	12.73	19.02
	IOG.L	-36.36	-78.29	-7.84
	ORBsup.R	18.49	48.1	-14.02
	TPOsup.L	-39.88	15.14	-20.18
	PCUN.R	9.98	-56.05	43.77
	AMYG.L	-23.27	-0.67	-17.14
	OLF.R	10.43	15.91	-11.26
	REC.R	8.35	35.64	-18.04
	PUT.R	27.78	4.91	2.46
	PCL.L	-7.63	-25.36	70.07
	CUN.R	13.51	-79.36	28.23
	PreCG.R	41.37	-8.21	52.09
	ITG.R	53.69	-31.07	-22.32
	AMYG.R	27.32	0.64	-17.5
	IFGtriang.R	50.33	30.16	14.17
weaken	ORBsupmed.R	8.16	51.67	-7.13
	LING.L	-14.62	-67.56	-4.63
	MTG.R	57.47	-37.23	-1.47
	HIP.L	-25.03	-20.74	-10.13
	ACG.R	8.46	37.01	15.84
	MFG.L	-33.43	32.73	35.46
	TPOmid.L	-36.32	14.59	-34.08
	ORBsupmed.L	-5.17	54.06	-7.4
	IOG.R	38.16	-81.99	-7.61
	ANG.L	-44.14	-60.82	35.59
	IFGtriang.L	-45.58	29.91	13.99
	MOG.L	-32.39	-80.73	16.11
	IPL.R	46.46	-46.29	49.54
	TPOsup.R	48.25	14.75	-16.86
	OLF.L	-8.06	15.05	-11.46

Our goal is to maximize the information lower bound:

$$\begin{aligned} IB &= \int p(A_{ij}, M_{ij}) \log q(M_{ij} | A_{ij}) dA_{ij} dM_{ij} \\ &\quad - \beta \int p(A_{ij}, W_{ij}) \log \frac{p(A_{ij} | W_{ij})}{r(A_{ij})} dA_{ij} dW_{ij}. \end{aligned} \quad (\text{A.11})$$

However, optimizing (A.11) is challenging, so we estimate the empirical distribution $p(M_{ij}, W_{ij}) = \frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N \delta_{M_{ij}}(M) \delta_{W_{ij}}(W)$. Eq. (A.11) then becomes:

$$\begin{aligned} IB &= \frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N \int p(A | M_{ij}) \log q(M_{ij} | A) \\ &\quad - \beta \int p(A | W_{ij}) \log \frac{p(A | W_{ij})}{r(A)} dA. \end{aligned} \quad (\text{A.12})$$

Thus, the entire information lower bound function becomes an integral over A . By using the reparameterization trick, we rewrite $\int p(A | M_{ij}) dA$ as $\int p(\epsilon) d\epsilon$ to facilitate gradient updates, where $\epsilon \sim \mathcal{N}(0, I)$.

Table B.11

Coordinates of identified ROIs in MDD dataset.

Type	ROI	x	y	z
strengthen	MOG.L	-32.39	-80.73	16.11
	MCG.R	8.02	-8.83	39.79
	MCG.L	-5.48	-14.92	41.57
	MTG.L	-55.52	-33.8	-2.2
	PCUN.L	-7.24	-56.07	48.01
	PreCG.L	-38.65	-5.68	50.94
	SPL.L	-23.45	-59.56	58.96
	SOG.R	24.29	-80.85	30.59
	MFG.R	37.59	33.06	34.04
	CUN.R	13.51	-79.36	28.23
weaken	PoCG.R	41.43	-25.49	52.55
	SMG.L	-55.79	-33.64	30.45
	PCG.R	7.44	-41.81	21.87
	ANG.L	-44.14	-60.82	35.59
	SFGmed.R	9.1	50.84	30.22
	AMYG.L	-23.27	-0.67	-17.14
	REC.R	8.35	35.64	-18.04
	CAU.R	14.84	12.07	9.42
	CAU.L	-11.46	11	9.24

Table B.12

Coordinates of identified ROIs in BD dataset.

Type	ROI	x	y	z
strengthen	MTG.L	-55.52	-33.8	-2.2
	MTG.R	57.47	-37.23	-1.47
	ITG.R	53.69	-31.07	-22.32
	TPOsup.R	48.25	14.75	-16.86
	MFG.L	-33.43	32.73	35.46
	SFGdor.R	21.9	31.12	43.82
	STG.R	58.15	-21.78	6.8
	MOG.L	-32.39	-80.73	16.11
	MCG.L	-5.48	-14.92	41.57
	PCUN.L	-7.24	-56.07	48.01
weaken	STG.L	-53.16	-20.68	7.13
	MOG.R	37.39	-79.7	19.42
	ACG.R	8.46	37.01	15.84
	INS.L	-35.13	6.65	3.44
	HIP.R	29.23	-19.78	-10.33
	HES.R	45.86	-17.15	10.41
	THA.L	-10.85	-17.56	7.98
	OLF.L	-8.06	15.05	-11.46
	SMG.R	57.61	-31.5	34.48
	SPL.R	26.11	-59.18	62.06
strengthen	PCL.R	7.48	-31.59	68.09
	HIP.L	-25.03	-20.74	-10.13
	THA.R	13	-17.55	8.09
	PCL.L	-7.63	-25.36	70.07
	PCG.L	-4.85	-42.92	24.67
	PCG.R	7.44	-41.81	21.87
	CAU.L	-11.46	11	9.24

Eq. (A.12) then becomes the final objective function:

$$IB = \frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N \mathbb{E}_{\epsilon \sim p(\epsilon)} [\log q(M_{ij} | f(W_{ij}, \epsilon))] - \beta D_{KL}(p(A | W_{ij}) \| r(A_{ij})). \quad (A.13)$$

The goal of our algorithm is to maximize IB , thereby preserving essential edge weight information and learning an adjacency matrix that is most relevant to the node information.

A.1. Hyperparameter settings

We present a performance bar chart for the hyperparameter experiments in Fig. A.5. The two hyperparameters are the coefficients λ_{CE} and λ_{KL} for L_{CE} and L_{KL} , respectively. The range of λ_{CE} is set to [1, 2, 3], and λ_{KL} to [0.001, 0.002, 0.003], as the value of L_{KL} is significantly large. It can be observed that the actual range of variation

across the three datasets is minimal with changes in hyperparameters, indicating that our method maintains stability against hyperparameter variations.

Appendix B. Information of identified ROIs

We present the three-dimensional coordinates of identified ROIs with functional connectivities have significantly strengthen or weaken by our model according to the MNI coordinate system. The results are summarized in Tables B.10–B.12

Appendix C. Availability of data and codes

The source code and datasets used in this study are available upon request. For access to these materials, please contact the corresponding author.

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